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Estimating colony strength and pollination efficiency in honey bees using a novel dataset and deep learning-based models

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ABSTRACT

This research introduces a deep learning-based computer vision system for automated honey bee identification and categorization at hive entrances to estimate different castes. The presence of pollen foragers at hive entrances is used to estimate colony health, colony strength and pollination efficiency. However, no study has created a tool for monitoring completely unmodified hives in real-world conditions. The system was trained and evaluated on video frames captured over two years via GoPro cameras, which encompasses various lighting and weather conditions. Bees were annotated to three categories: 1. Worker bee – No pollen, 2. Worker bee – pollen and 3. Drone bee. Five object detection models were compared: YOLOv10, FrCNN, RetinaNet, FCOS, and SSD. While YOLOv10 (93.9 % accuracy, 0.92 F1), FrCNN (93.4 %, 0.91), FCOS (92.3 %, 0.88), and RetinaNet (91.8 %, 0.9) achieved excellent results, SSD (54.2 %, 0.69) showed average performance. Best mAP scores were achieved with FrCNN and Friedman's test statistically confirmed a significant difference in model performances (Friedman's statistics = 20.6667, $p < 0.001$). The proposed system provides a noninvasive, easy-to-use, and cost-effective method for monitoring honey bee behavior by providing valuable data for beekeepers and researchers. The study also contributes to a publicly available dataset of 4,590 frames with 79,212 bee annotations.

1. Introduction

The value of crop pollination and honey production in the United States by honey bees are respectively estimated to be around $17 billion [1] and $4.74 billion [2]. Globally honey production was approximately 1,894,000 tonnes in 2025 [3] with natural honey valued over $7 billion [4] and beeswax valued over $51 million in 2024 [4]. In commercial cropping systems, healthy honey bee colonies stocked at the right densities can improve pollination efficiency [5]. This in turn has a larger impact on the revenues generated by both the growers and the beekeepers. Loss of pollinators, especially honey bees, may be significantly detrimental to the agricultural economy. In addition, it may affect crop production, food security, and overall ecosystem stability [6,7]. Poor nutrition, *Varroa* mites, pathogens, and the impacts of pesticides are among several factors that have been cited as probable causes for honey bee colony declines, although further research is needed to fully comprehend their impacts. In recent times, other bacterial diseases such as European foulbrood (EFB) have also caused significant impacts on colonies [8]. This is why it is extremely crucial that researchers and stakeholders can assess colony health by estimating the population size of a colony and understand its population dynamics.

The honey bee colony has three castes: queens (fertile females), workers (sterile females), and drones (males). Workers dutifully perform all tasks in the colony (including foraging for pollen, nectar and water [9]), the queen lays the eggs, and the drones mate with virgin queens (usually during the spring). While honey bee foragers are collecting

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resources from flowers, they also successfully pollinate by transferring pollen from the anthers to the stigma of the flower. Pollination from honey bees can be estimated by counting the number of returning forager honey bees with pollen pellets on their hind legs. To measure pollination success, both the strength and number of honey bee colonies play a significant role in addition to the number of returning pollen foragers. The number and type of returning forager can be used to estimate colony strength. A significant number of returning pollen foragers indicates that the colony is actively rearing brood [10]. Researchers have determined that colonies with more than 100 incoming foraging bees per minute can be considered good pollination colonies[8]. However, weather and temperatures can largely influence foraging. To estimate colony strength using forager behaviors, temperatures should ideally stay above 65°F (18°C) or above, and windspeed should remain below 10 miles per hour (16 km/h)[8].

For decades, many researchers have attempted to successfully measure colony strength through different ways. The most common method involves opening a colony and estimating its population by counting the percentage of adults and brood. However, this strategy alone does not account for foragers, which are searching for pollen and nectar outside the colony hive box. Researchers have attempted to account for foragers through several methods. While some have attempted to build forager traps to count pollen foragers as they enter the colony [11], others have tried either standing by the colony and manually counting incoming pollen foragers or recording short videos and later reviewing them to count the pollen foragers frame by frame [12]. Other methods used in the past included using infrared sensors [13], harmonic radar [14] and radio frequency identification (RFID) tags [15–17]. Installing a trap is cumbersome and often may impose additional stress on the foragers. The count may also not be accurate given that only a few foragers can enter the colony through the trap at a time (reduced forager entrance benefits the observer who must count manually). In all the video recording methods, it takes hours, weeks, or even months to manually review and distinguish pollen from nectar foragers, depending on how many colonies were recorded, the duration of the recording, and how many video frames were captured. Any of the methods involving infrared sensors, harmonic radars or RFID are often not easily reproducible for stakeholders and researchers due to lack of expertise, resources and a high cost.

Deep learning is a subset of machine learning that involves training artificial neural networks with multiple automated feature extraction layers known as deep neural networks. Deep learning has significantly changed the field of computer vision where essential features in imagery are extracted using a series of filters which is defined by a mathematical operation known as convolution. Since a digital image is a two-dimensional signal, a 2D convolution filter (or a kernel) is used to process (convolve) the input image. The filter slides over the image and performs a convolution operation, which involves element-wise multiplication and summation. Applying a 2D convolution filter over the image creates a new matrix called the feature map that represents the spatial presence of the detected features. By stacking multiple convolutional layers, Convolutional Neural Networks (CNNs) automatically learn to detect and extract hierarchical features from raw image data, such as edges, textures, and more complex patterns, achieving outstanding results on several complex cognitive tasks, matching or even beating those provided by human performance [18].

Image classification and object detection are the fundamental cores of computer vision. Image classification identifies and categorizes images or specific regions within an image and assigns a label to the object. Object detection extends image classification by identifying multiple objects present in the image and locating their positions in the form of bounding boxes (rectangular boxes represented by coordinates of its corners, or coordinates of its center along with height and width). Before the rise of deep learning, object detection relied heavily on feature engineering with handcrafted features using Histogram of Oriented Gradients (HOG) [19] and Scale-Invariant Feature Transform (SIFT) [20]. Traditional machine learning approaches include methods like comparing geometric properties of objects or feature matching for object detection [21]. However, these methods had limitations in terms of accuracy, detection speed, and computational complexity. With the introduction of CNNs, the fields of image classification and object detection saw huge improvements. Deep learning object detection models provide a scalable, accurate, and automatic real-time monitoring system for the health and activity of honey bee colonies, along with their non-intrusive nature, which minimizes the stress on honey bees and frequent colony disturbances.

Deep learning-based object detectors are mainly classified into one of two categories: one-shot detectors, and two-shot detectors [22]. One shot detectors predict the bounding box and class probability in a single pass across an image. Class probability is the likelihood that a given image or region of image belongs to a specific class. This makes one shot detectors computationally efficient, but often less accurate than two shot detectors. Two shot detectors use two passes on an input image. The first pass proposes regions of interest (ROIs) representing potential object locations. The second pass refines the proposals and then classify the objects. Thus, they are more computationally expensive but often more accurate than one-shot detectors [22].

Over the last decade, various object detection and classification models have been proposed in literature. Most of the detectors are based on CNN architecture or backbone. In 2012, AlexNet, an early CNN architecture, was proposed [23]. It had an unparalleled performance on ImageNet 2012 Challenge, one of the most popular benchmarks for image classification. It significantly outperformed traditional machine learning-based approaches. Through its success, AlexNet showcased the strength of CNNs and deep learning [23]. In the years since, numerous other deep learning-based object detectors have capitalized on the success of AlexNet and other advancements in deep learning. Ren *et al.* proposed Faster Region-based CNN (FrCNN) [24], an advancement of the earlier Fast Region-based CNN (Fast R-CNN) [25] by including a trainable region proposal network. Since 2015, several versions of You Only Look Once (YOLO) models have been proposed [26,27] with the latest model being YOLO version 10 (YOLOv10) [28]. RetinaNet was proposed [29] with the introduction of focal loss to address class imbalance. Fully Convolutional One-State Object Detection (FCOS) was proposed [30] which simplifies the detection process by using a fully convolutional network to predict bounding boxes and class scores directly from feature maps, without the need for anchor boxes. Single Shot Multi-box Detector (SSD) [31] was proposed with a set of predefined anchor boxes of different aspect ratios and scales at each feature map location.

Deep learning-based object detection methods have become dominant in various application areas due to their high performance and efficiency, supported by CNNs. These methods are used across many fields, such as computer vision, healthcare, robotics, and agriculture. Deep learning object detection models provide a scalable, accurate, and automatic real-time monitoring system for the health and activity of honey bee colonies, along with their non-intrusive nature, which minimizes the stress on honey bees and frequent colony disturbances. In a recent study [32], a YOLOv5-based model was proposed for detecting pollen-bearing honey bees. The study included 2051 images with approximately 60,000 annotated bees, of which only 1758 were pollen-bearing foraging bees. However, the study required a two-level object detection and image classification model. The first model detected honey bees (with or without pollen), and the second model classified the detected subimages. Additionally, the study excluded drone bees found at the entrances of beehives.

Another study [33] demonstrated a deep learning-based system using a low-cost microcomputer and an off-the-shelf imaging system to detect, count, and differentiate forager bees with or without pollen. Even though this successfully attempted to automate forager counts, the system has several limitations. The software is not available freely and must be purchased, the colony used is not representative of typical

healthy colonies, the algorithm does not differentiate between workers and drones, it does not account for changing light intensities, and the setup is also not convenient for an average researcher or a stakeholder. Other ad-hoc approaches to bee detection and counting [34,35] have been proposed recently, but they likewise: (1) fail to differentiate between drone and worker honey bees, (2) do not consider changing light intensities, and (3) do not provide openly available software or datasets for use by other researchers and stakeholders.

Although many studies have demonstrated feasibility of object detection for honey bee colonies, these have involved altering the colony's entrance in such a way as to lessen variability and increase model performance. However, such alterations reduce a model's capability to infer in high-noise situations and require additional effort on the part of the research team to fit the data with the model. These factors limit utility and create barriers to the usage of a machine learning system by beekeepers, growers and researchers. Unlike previous studies that required modifications to hive entrances or controlled conditions, our study developed a fully non-invasive, robust real-time and real-world deep learning monitoring system for honey bees. To create a model capable of functioning within a noisy and realistic environment, it was first necessary to generate a high-quality dataset from these real-life conditions. With all these factors in mind, there was a need to create an automated solution that both stakeholders and researchers can use to monitor honey bee colony foraging patterns. We collected data over two years, capturing a broad range of environmental factors, making our study unique in robustness and practical applications. Thus, this study is unique for three reasons. (1) GoPro-style camera was used to create a robust, representative and extensive high-quality dataset comprising of frames from video recordings with changes in weather parameters, time of day, and light intensity. This allowed the trained models to perform reliably in real-world settings. Also, GoPro cameras are readily available in the market and relatively easy to use. (2) A publicly available dataset classifying honey bees into three distinct classes (worker bees (WB) no pollen, WB pollen, and drone bees) has been created through this study. The dataset was carefully annotated by a human subject matter expert for training deep-learning-based models. (3) The efficacy of various deep learning object detection models to detect honey bees (while also accurately classifying them into the three classes) was also examined thoroughly. The following models were optimized, trained and tested on the open access dataset: (i) YOLOv10, (ii) FrCNN, (iii) SSD, (iv) FCOS and (v) RetinaNet.

2. Materials and methods

2.1. Experimental datasets

The experimental datasets for this study were created *de novo* using videos collected within the research apiary on Mississippi State University's campus in Starkville, MS, USA. Colonies were selected for recording based on cursory estimates of worker bee and pollen forager abundance, but colony strength varied between hives and time of recording. Creating the dataset involved three key steps: (1) collecting an array of videos across different times of year, weather, and lighting conditions, (2) video processing, and (3) image annotation. To coordinate these activities and quantify the diversity of conditions, all important information was recorded in a centralized spreadsheet.

No attempt was made to differentiate between behavioral groupings of worker bees without pollen because they are physically indistinct and therefore cannot be separated by image data alone. Hypothetically, these groups might be separated using a tracker monitoring object position to determine behavioral groups, or worker bees without pollen could be excluded from detection by the algorithm. Mating swarms are a rare but key event in the colony life cycle. Likewise, drone bees are only present for brief windows of time at hive entrances. This contributes to significant underrepresentation compared to other targeted classes. Therefore, videos containing mating swarm events or drone bees were preferentially targeted for annotation to increase representation of drone bees and more unusual climatic or lighting conditions.

2.2. Data acquisition and image annotation

The summary of the data development process is shown in Fig. 1.

2.2.1. Timing of recording and hive choice

Multiple studies have demonstrated that honey bee colony foraging activity is greatly impacted by variations in weather and sunlight [36, 37]. To increase the diversity of weather conditions, recordings were taken in the morning, at midday, and in the evening during a two-year period. Temperature and solar radiation, a quantification of the energy provided by the sun over a given area, have been demonstrated to predict 78 % of bee foraging [38]. As we desired to create a dataset representative of conditions at different times of the year and in different conditions, it was necessary to record weather information for every video. An AcuRite 5-in1 Weather Sensor [39] was installed within the apiary. At the time of each recording, temperature, humidity, wind speed, and wind direction information were typically logged from the weather station. However, seven videos were not immediately logged. These were noted and historical records from a NRCS National Water and Climate Center (NWCC) station on campus [40] were used to provide weather information. As our station also could estimate solar radiation, NWCC records were used to estimate average hourly solar insolation for our videos.

Before camera placement, colonies were checked for brood presence and visually assessed at hive entrances. Initially, hives were targeted for video recording if workers were relatively abundant. However, as dataset creation progressed, worker bees without pollen became increasingly overrepresented. To counter this, a high relative proportion of pollen-bearing bees at hive entrances was used to select colonies beginning in February 2023.

2.2.2. Recording equipment

Two GoPro Hero cameras (a GoPro Hero 9 and a GoPro Hero 11) were mounted on standard honey bee colony boxes facing the colony entrances. Using their built-in displays, cameras were positioned such that about the bottom third of the field of view was occupied by the colony entrance. Cameras were mounted on colonies using an adhesive-based mount or two chest harnesses secured around the box (Fig.. 1a, 1b). To reduce data burden, each video was only recorded for 30 – 60 minutes at a frame rate of one or two frames per second (fps). All videos were recorded in 4K resolution. Video quality was similar between GoPro Hero 9 and GoPro Hero 11 recordings. Three videos were recorded at 30 fps to capture mating swarm activity, a rare event. Video metadata was logged for each recording.

2.2.3. Video pre-processing

Once videos were collected and logged, they were uploaded to Mississippi State University OneDrive and indexed according to date and time. Video files were reviewed to identify the relative abundance of pollen foragers compared to workers without pollen (guard bees, incoming nectar foragers), as well as to identify if drone bees were present. Videos with drone bees or highly abundant pollen foragers were considered more important to the dataset and given priority for labeling. Matlab [41] was used to extract video frames into 4K images and crop them to the bottom third of the image. By cropping the field of view to only the hive entrance, extraneous data and complex, confounding backgrounds were removed.

2.2.4. Choosing and differentiating annotation classes

Honey bee workers are the only caste that forages in a honey bee colony. Individuals of this caste have hind leg corbiculae, also called "pollen baskets", which are used for packing pollen during foraging. For our purposes, these "pollen baskets" allow workers with pollen to be



Fig. 1. The dataset development process. (a) Left to right: Video recording with an adhesive-based mount, two chest harnesses, and an uncropped video frame. (**b**) After recording, frames were extracted and cropped to the bottom third using Matlab [41]. (**c**) An annotated frame created using Labelme [43].

distinguished from workers without pollen (such as guard honey bees or nectar foragers). At colony entrances, drones are also sometimes present. These bees are morphologically distinct from worker bees. The amount of pollen entering a colony can indicate the presence of brood, a change in activity at colony entrances may reflect colony health, and drone bees are only present at entrances before mating flights [42]. Therefore, three classes outlining these important groups were used: worker bees with pollen, worker bees without pollen, and drone bees (Fig. 1c).

2.2.5. *Annotation process*

Labelme [43], an open-source annotation software, was used to create simple, axis-aligned bounding box annotations for every bee within an image. Annotations were saved in the Labelme JSON format. During labeling, a reviewer familiar with honey bee morphology identified and assigned individuals annotations. The reviewer assigned it to one of the three classes, “WB without pollen”, “WB with pollen”, and “drone bees” where WB indicated worker honey bees. In many cases, bees were partially occluded either by other bees or at the edge of the image. As this case is exceedingly common at hive entrances, the occluded individuals were annotated and identified to class. If partially occluded bees were not evidently drones, or their hind legs were not visible (and thus it was impossible to identify whether they bore pollen), occluded individuals were assigned as worker bees without pollen, In addition to occlusion, many individual bees were blurry due to speed of bee flight and camera shutter speed. However, many of these blurry bees nonetheless clearly bore pollen. Therefore, blurry objects were able to be correctly identified to class.

Due to the large number of videos recorded, the presence of up to hundreds of individual bees within a single frame, and a limited number of annotators, it was not possible to use inter-annotator agreement to ensure dataset quality. To account for this limitation as well as retain quality, frames were reviewed several times at different points in the process by an expert in honey bee morphology. Annotation consistency was ensured by manual review at the time of and following image annotation. Following annotation but before newly annotated images were added to the full dataset, they were again reviewed to ensure class and label accuracy. When creating new versions or remapping classes for analysis, the entire labeled dataset was reviewed a third time to eliminate error and increase annotation consistency. Typos in class names were corrected, and bounding boxes were tightened or removed if inaccurate, and inappropriate classifications were relabeled.

2.3. *Deep learning methodology*

In this study, we distinguish between pollen foragers and non-pollen foragers by examining the presence of pollen pellets on the hind legs of honey bees. Our objective was to accurately detect and classify bees that carry pollen compared to those that do not, based on the presence of these pellets. However, the average size of the pollen pellets is only 15 by 15 pixels, which complicates the classification process due to their small size.

We have chosen four deep learning object detection models that are commonly utilized in honey bee research. The YOLO models (versions

v3, v4, v5, and v7) are particularly favored for bee-related detection tasks due to their balance of speed and accuracy. These YOLO models are employed for a variety of applications, including detecting beehives [44], bees [45], monitoring bee entrances [46], identifying *Varroa* mites [45,47], and tracking pollen [33,47]. Faster R-CNN models were employed for detecting larvae, bees, and mites [46,48] SSD was also used for bees, *Varroa* mites, and bee swarming detection [47,49]. RetinaNet has been used for multi-species and mite detection [47,50], and bumble bee flight tracking [51]. In addition to the commonly used four models, we employed the FCOS model because it is anchor-free, consumes less memory, and features a simpler architecture usage, and has a simpler architecture [30]. The following subsections explain the architecture and advancements of the object detection models in detail that were used in this study.

2.3.1. YOLOv10X

YOLO presented a new approach to object detection. Instead of traditional two shot detectors where regions were generated and then classified as objects, YOLO proposed a single neural network prediction bounding boxes and class probabilities directly from full images in one evaluation [28]. Subsequent versions of YOLO improved accuracy while retaining high inference speed with YOLOv10 pushing its boundaries with holistic efficiency-accuracy driven model design strategy. YOLOv10 achieved state-of-the-art performance with significantly reduced computational overhead with enhanced version of CSPNet (Cross Stage Partial Network) [52] as the backbone network, responsible for feature extraction.

The different architecture models and labeled classes used in the dataset are shown in Fig. 2.

2.3.2. FrCNN

FrCNN is an advancement of R-CNN architecture. FrCNN added a Region Proposal Network (RPN) to identify possible location of objects. The RPN is also trainable and is merged with Fast R-CNN architecture into a single network by sharing their convolutional features which significantly reduced the computation time required. FrCNN consists of a backbone network for feature extraction, a Region Proposal Network (RPN) for proposing candidate object regions, and a second stage for refining these proposals and classifying them [24]. FrCNN is known for its high accuracy and ability to detect small objects and densely packed objects. We trained this model on our bee dataset with ResNet50 [53] backbone pretrained on ImageNet dataset using AdamW optimizer, instead of Visual Geometry Group-16 (VGG-16) backbone mentioned in the FrCNN paper. Residual Network (ResNet) introduced skip connections or residual connections, allowing for the training of much deeper networks without suffering from the vanishing gradient problem and increases the performance of the model.

2.3.3. RetinaNet

RetinaNet [29] is a single shot object detector improved upon existing one shot detectors using focal loss and FPN. Focal loss addresses class imbalance during the training, which in our study, was essential

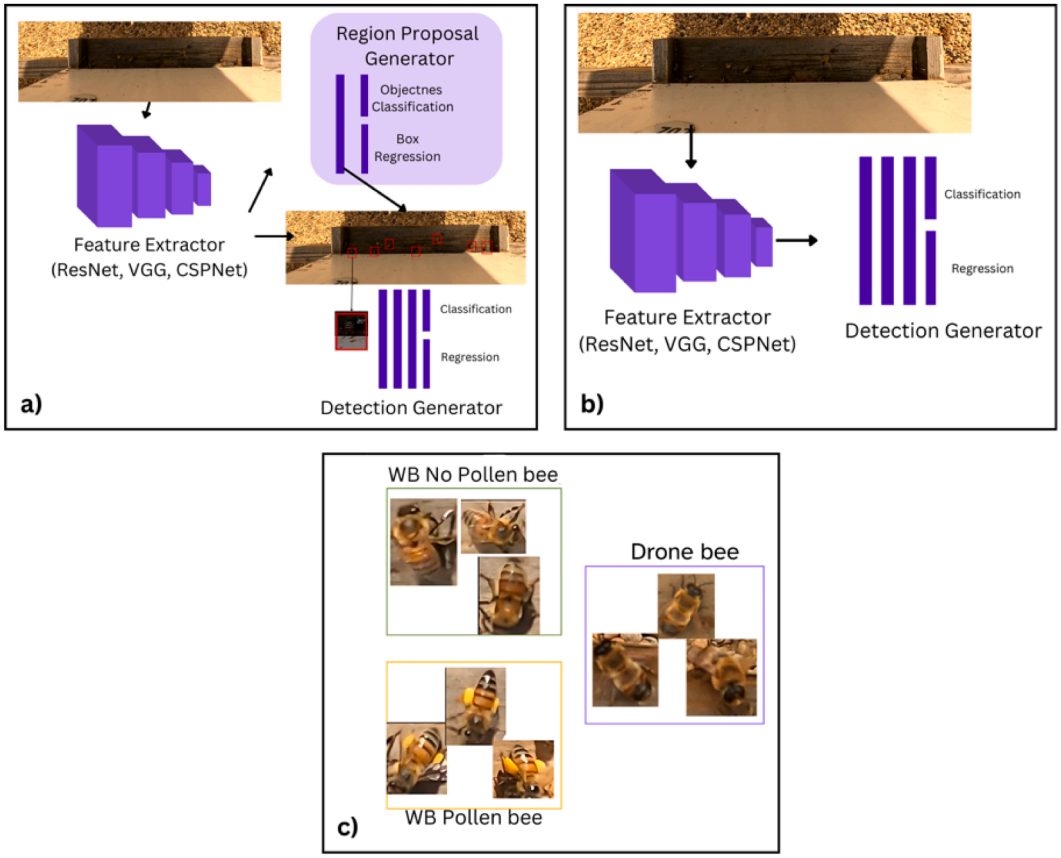


Fig. 2. Architectural frameworks and target honey bee classes used for model training. (a) Two-shot detector architecture that illustrates the multi-stage pipeline where a feature extractor identifies high-level features before a Region Proposal Generator proposes potential object locations for final classification and bounding box regression. (b) One-shot detector architecture that shows the streamlined, single-pass process where the detection generator predicts class probabilities and bounding box coordinates directly from the feature maps (c) Class Differentiation through visual comparison of the three annotated categories - WB No Pollen (green outline), WB Pollen (yellow outline highlighting the distinct yellow/orange pellets on the hind legs), and Drone bee (purple outline highlighting larger body size and eye structure). Red and colored rectangular overlays represent the bounding boxes used to define the spatial coordinates and extent of each bee against the hive entrance background. WB indicates worker honey bees.

with 95 % of our dataset containing a single class. FPN enhances multi-scale feature representation by combining low-resolution, semantically strong features with high-resolution, semantically weak features and this improved the performance in detecting objects. We used ResNet-50 backbone pretrained on ImageNet dataset for feature extraction and used AdamW optimizer to train RetinaNet model on our custom dataset.

2.3.4. FCOS

FCOS is a fully convolutional anchor-free as well as proposal free one-shot object detection model. Removing the need for anchor boxes reduces the complexity and computation related to anchor boxes which simplifies detection, leading to faster training and detection and works well even for small object detection. FCOS also avoids all hyper-parameters related to anchor boxes, which are often very sensitive to the final detection performance [30]. We used ResNet-50 backbone pretrained on ImageNet dataset for feature extraction and used AdamW optimizer to train our FCOS model on the bee dataset.

2.3.5. SSD

SSD [31] is a one-shot object detector with a backbone and SSD head. We used VGG16 [54] as the backbone of the network for feature extraction. SSD discretizes the output space of bounding boxes into a set of default boxes over different aspect ratios and scales per feature map location. At prediction time, the network generates scores for the presence of each object category in each default box and produces adjustments to the box to better match the object shape [31]. It also uses multi-scale feature maps for detecting objects at different scales.

2.4. Experimental setup

All the object detection models in this study were trained with a batch size of four on a Tesla P100 GPU with 16GB of VRAM and 30GB RAM. The full configuration and training parameters used in our models are described in Table 1. All models accept input images of any dimensions but YOLOv10 automatically transforms all input images to input size of 1080 × 1080 and SSD automatically transforms the images to 1088 × 1088 before passing to the backbone. Early stopping was used while training, i.e. if the models did not improve their accuracy within the next ten epochs, the best model was saved, and training was exited. The initial learning rate was carefully chosen for all models using random search. Using small initial learning rate in our models is effective because the backbone networks have already learned useful features from large datasets like ImageNet or Common Object in Context (COCO). A learning rate scheduler was used with the one cycle learning rate policy, potentially leading to faster convergence and better model performance. For every epoch, the dataset was augmented with a variety of random transformations applied to all images and the bounding boxes changing accordingly. Using different augmentations for every other epoch results in models training on a diverse dataset, enhancing its ability to generalize well while effectively reducing risk of overfitting. Motion blur and Gaussian blur were used with maximum kernel size of (3, 7) and probability of 0.25. Random brightness and contrast were used with brightness and contrast limit of 0.25 (one being the highest) with guaranteed probability. The dataset would randomly flip images horizontally with 0.4 probability and rotate images with the same probability. Additionally, a random part of the input was cropped without loss of bounding boxes with probability of 0.3. The larger, more diverse and representative (with respect to the distribution of the target dataset) the training data, the more effective the deep learning model performs on unobserved data. These augmentation methods provide complementary benefits in computer vision applications [55]. Other hyperparameters such as augmentation probabilities and optimizer configurations didn't have significant impact in performance with the change in hyperparameters. The optimizer configuration was set to the PyTorch's default values widely validated in deep learning applications. This was more effective and relevant in our experiment. The augmentation probabilities and other hyperparameters were chosen based on empirical validation or set to default values.

2.5. Model evaluation

To evaluate the proposed models, the dataset was split into train (60 %), validation (20 %) and test sets (20 %) for all models. The validation dataset acted as an independent set of data the model had not been trained on allowing for unbiased performance while training. Additionally, the validation dataset helped in selecting the best-performing model among different iterations or configurations. While training, after every epoch (complete cycle through the dataset), the model was evaluated on validation dataset. The same losses that were calculated on the train set were also calculated for this. Additionally, the models were assessed on different metrics:

Intersection over Union (IoU): A metric that evaluated the overlap between the predicted bounding box and the ground-truth bounding box. It matches the predicted bounding box to the ground truth bounding box when it is above a certain threshold (in our models, we used IoU=0.5). IoU is used to determine whether a predicted bounding box is considered a true positive. The following formulae (1 – 4) were used to evaluate the different parameters as shown.

Accuracy: Accuracy measures the overall correctness of the predictions made by the model. In object detection, we measure accuracy and other metrics by matching the predicted bounding boxes with the ground truth bounding box using IoU. Accuracy refers to the ratio of the predicted bounding boxes overlapping with the ground-truth bounding boxes correctly classifying them to total instance of bounding boxes.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}. \quad (1)$$

Precision: Precision is the ratio of true positive predictions to the total number of positive predictions (true positives and false positives). It measures how many of the detected objects are correct. High precision means when a model predicts an object, it is very likely to be correct.

$$Precision = \frac{TP}{TP + FP}. \quad (2)$$

Recall: Recall is the ratio of true positives to the total number of actual positives (true positives and false negatives). It measures how well the model calculates all relevant objects from the frame. High recall ensures the model captures most of the objects to be detected in the frame, even at cost of false positive.

Table 1
Configuration and training parameters of the deep learning-based object detection models trained. FLOPS indicates floating-point operations per second and FPS indicates frames per second.

Models	# Parameters (M)	Epochs Trained	Backbone Network	Pretrained Dataset	Optimizer	Initial Learning Rate	FLOPS (G)	FPS
YOLOv10	31.6	45	Enhanced CSPNet	COCO	SGD	$1.0e^{-2}$	169.8	25.93
FrCNN	43.3	47	RestNet50	ImageNet	AdamW	$6.0e^{-5}$	280.37	10.98
RetinaNet	36.4	58	ResNet50	ImageNet	AdamW	$9.0e^{-5}$	152.24	22.2
FCOS	32.1	43	ResNet50	ImageNet	AdamW	$1.5e^{-4}$	128.21	21.77
SSD	24.0	79	VGG16	ImageNet	AdamW	$3.0e^{-3}$	98.20	23.72

$$Recall = \frac{TP}{TP + FN} \tag{3}$$

F1 Score: F1 Score is the harmonic mean of precision and recall, providing a balance between the two. The F1 score provides a single metric that reflects both precision and recall, when there is a need for balance between precision and recall as they are inversely related.

$$F1\ Score = 2 * \frac{Precision * Recall}{Precision + Recall} \tag{4}$$

In these formulas (1–4), TP indicates True Positive (correctly predicted positive cases), TN indicates True Negative (correctly predicted negative cases), FN indicates False Negative (incorrectly predicted as negative), and FP indicates False Positive (incorrectly predicted as positive).

mAP at IoU Threshold 0.5 (mAP@50): mAP@50 is a metric used in object detection to evaluate models based on their precision and recall at fixed IoU threshold of 0.5. mAP@50 is calculated by averaging Average Precision (AP) values across all classes. AP is a metric that combines precision and recall. It is the area under precision-recall curve, which plots precision against recall at different confidence (conf.) threshold. mAP@50 is a common metric used in object detection that assess the model's performance in detecting object.

mAP across IoU Thresholds from 0.5 to 0.95 (mAP@50-95): It is a more comprehensive metric that evaluates object detecting model across several IoU threshold (0.5 to 0.95, with interval of 0.05). By averaging AP scores at different threshold, it provides a better metric for model comparison.

The confidence score for each prediction is derived from the product of the box-localization probability and the class-specific probability. In the YOLOv10 architecture, this is calculated using formula 5, where the first term represents the model's certainty that a bounding box contains an object and the second represents the conditional probability of the assigned class. For the two-shot Faster R-CNN (FrCNN) model, the score is generated by the Softmax layer of the classification head after region proposals are refined.

$$Confidance\ Score\ =\ P(Object)\ \times\ P(Class|Object) \tag{5}$$

2.6. Statistical analysis

A Friedman's statistical test was conducted to compare the performance of multiple multiclass classifiers [56]. The null hypothesis stated that there was no significant difference in the average performance of the classifiers, while the alternative hypothesis stated that at least one classifier performed significantly differently. The Cohen's Kappa scores for the tested models were computed from the confusion matrix that provide a measure of agreement between the models' predictions and the ground truth annotations [56]. All statistics were done using Python programming language with the *scipy* library for the Friedman's test and *scikit-learn* for calculating Cohen's Kappa scores.

3. Results

3.1. Dataset and annotation statistics

A labeled dataset consisting of 4,590 frames distributed across 15 different video recordings representing diverse lighting and seasonal conditions in Starkville, MS, USA was developed [57]. Across all images, 79,212 annotations were created belonging to one of the three classes: WB no pollen, WB pollen, and drone bees. Dataset distribution was heavily skewed toward WB no pollen, with only 5.4 % of annotations representing WB pollen and 0.7 % representing drone bees (Table 2). The distributions of classes and images across videos represented in the dataset are described in Table 2. The high number of annotations relative to labeled frames in specific instances is due to the high density of honey bee activity on certain days at the hive entrance during peak periods. In these frames, bees often appear densely packed or partially overlapping, requiring individual annotation to capture the full extent of the colony's entrance traffic. We specifically choose those frames for annotation to include high bee density and to develop a model capable of functioning within the noisy and realistic environments encountered in unmodified hive entrances.

Table 2
Class and image distribution across the labeled dataset. Due to the nature of honey bee biology and seasonality, classes were heavily skewed toward worker bees (WB) no pollen.

Date and Time Recorded	Labeled Frames	Percent Representation	Total annotations		
			WB No Pollen	WB Pollen	Drone Bees
Mar 1st, 2023, 3:37 PM	431	9.39 %	0	0	0
Jul 11th, 2022, 9:00 AM	1,368	29.80 %	39,417	297	0
Aug 1st, 2022, 8:46 AM	25	0.54 %	963	16	0
Sep 28th, 2022, 12:00 PM	13	0.28 %	163	6	0
Oct 10th, 2022, 11:00 AM	1	0.02 %	103	0	0
Oct 10th, 2022, 1:00 PM	638	13.90 %	12,850	589	19
Feb 7th, 2023, 1:40 PM	1,279	27.86 %	3,947	2,771	0
Mar 22nd, 2023, 2:40 PM	249	5.42 %	4,530	216	313
Mar 22nd, 2023, 3:30 PM	116	2.53 %	2,954	67	227
Aug 22nd, 2023, 8:35 AM	50	1.09 %	721	7	0
Sept 5th, 2023, 8:35 AM	3	0.07 %	222	1	0
Sept 20th, 2023, 9:50 AM	200	4.36 %	4,043	150	0
Oct 12th, 2022, 11:38 AM	126	2.75 %	2,351	74	0
Aug 16th, 2023, 9:55 AM	1	0.02 %	26	0	0
Sept 7th, 2023, 10:20 AM	10	0.22 %	1,249	25	0
Total in Dataset	4,590	100 %	73,981 (~94 %)	4,272 (5.4 %)	559 (0.7 %)

Drone bees are present only during certain times of the year and are rarely observed at colony entrances. As a result, there was barely any representation in the dataset compared with the other two classes. Pollen-bearing workers represent a single behavioral group, whereas WBs without pollen constitute several behavioral groups, including nectar foragers and guard bees. Thus, only 5.4 % of the annotations represent pollen-bearing bees, whereas 94 % are bees without pollen.

3.2. Model training, validation, and testing

Well-labeled datasets are the foundation of deep learning model development. During training, the model learns to identify objects by

iteratively adjusting millions of parameters (weights and biases of the nodes, also called the neurons) to minimize the difference between the ground truth in the dataset and predictions. The predictions are the bounding box and class labels. To minimize the difference, we calculated the loss between ground truth and prediction separately for classification task (cls loss) and bounding box (box loss). The two losses were then combined as total loss in all models except YOLOv10 and the models used an optimization technique called backpropagation, in our models Adam or SGD, to reduce the losses in subsequent iterations (Fig. 3). YOLOv10 uses distribution focal loss (dfl loss) rather than box loss to refine the model's ability to predict bounding boxes by treating bounding boxes coordinates as a probability distribution.

We tested the trained models on a separate test set to assess their performance on new, unseen data. While the validation dataset is crucial for model tuning and development, the test set is important to provide an unbiased final evaluation of the model's performance. It ensures that the model generalizes well to new data and is ready for real-world deployment. The model then predicts the bounding boxes, its associated predicted class label and score. The score in prediction refers to a numerical value in range [0, 1] that indicates the confidence or probability of the predicted class. The metrics introduced in the model evaluation section were calculated for each individual class and plotted against different confidence threshold (score) (Figs. 4 and 5).

A confusion matrix is often used in object detection and classification to show how well the model's prediction matches the actual outcomes while accounting for multiple classes. The confusion matrix was thus also calculated on test datasets for an accurate and unbiased evaluation of the models (Fig. 6). Overall, these confusion matrices showed varying performance levels among our models with SSD showing weak performance, while the other four models displayed strong performance. The latter models could accurately distinguish between the castes of honey bees but occasionally struggled with distinguishing honey bees from background, likely due to the challenges posed by varying lighting conditions and rapid movements of the honey bees. Note that the Background class in the confusion matrices (Fig. 6) serve a specific evaluative purpose. They represent instances where the model either misidentified the background as a bee (False Positive) or failed to localize a bee entirely (False Negative). Because background was not a trained class, the diagonal cell for background is zero, as the focus of this metric is the correct classification of bee castes versus localization errors against the environment.

The Friedman's statistical test revealed significant differences in the performance of the multiclass classifiers being evaluated (Friedman statistics $= 20.6667$, $p < 0.001$). YOLOv10 achieved a Cohen's Kappa score of 0.6105, indicating substantial agreement. FRCNN, RetinaNet, and FCOS showed moderate agreement with scores of 0.5629, 0.4921, and 0.5152, respectively. In contrast, the SSD model exhibited a Cohen's Kappa score of -0.1391, suggesting agreement worse than chance. These results highlight that YOLOv10 had the strongest agreement with the annotations, followed by FRCNN, RetinaNet, and FCOS, while SSD performed poorly in terms of agreement.

As this study focused on vision-based models, we only used visual data to distinguish between our classes (pollen, non-pollen carrying bees and drone bees). It is more practical to implement in real-life scenarios without the need to collect environmental factors. Even though we collected the environmental factors data for our experiment, our approach is invariant to temperature, lighting conditions, wind speed, etc. Thus to assess the impacts of environmental factors, we analyzed model accuracy across different weather conditions. We classified frames intro 4 weather conditions: Sunny (595 frames), Mostly Cloudy (42 frames), Partly Cloudy (28 frames) and Cloudy (19 frames). The performance metrics for Sunny (93.9 % accuracy, 0.916 f1-score), Mostly Cloudy (92.9 % accuracy, 0.901 f1-score), Partly Cloudy (92.1 % accuracy, 0.896 f1-score) and Cloudy (98.8 % accuracy, 0.988 f1-score) indicate that the models are doing well across all weather conditions and are invariant to them.

4. Discussion

The diversity in our data collection addresses a critical gap in previous research, which often focused on limited conditions and led to overfitting or poor performance in a realistic environment. Moreover, the systematic logging of environmental factors (including temperature, wind direction and speed, humidity and solar UV radiation) provides a valuable context for analyzing the behavioral patterns of honey bees. Future studies can utilize this dataset to explore correlations between environmental conditions and foraging behavior.

In our study, we chose to use deep-learning based object detectors instead of two-stage pipeline combining two-stage detectors with classifiers to implement our system for bee detection and classification. For the two-stage pipeline method, an object detector would detect every honey bee in the image. Next, these detections would be cropped from the background and passed to a CNN classifier to identify the caste (worker or drone honey bee) and behavioral level (pollen-bearing versus non-bearing). While it allows for a more specialized classification, these systems require separate training of the detector and the classifier. Although this two-stage pipeline may resemble conventional two-shot detectors, they operate differently. A two-stage detector simultaneously trains both bounding box loss and classification loss using advanced techniques for improved classification. An example is using Feature Pyramid Network (FPN) to extract multi-scale features from input image which improves both detection accuracy and classification accuracy [58,59]. We used a two-stage pipeline to build two models for our experiment: the first model (object detector) focused solely on bounding box prediction, and the second model (CNN classifier) classified the honey bees by taking the results from the first model as input. This separation was time consuming in both training and testing and eventually errors in detection can propagate from the detector to classifier and create suboptimal performance. Thus, instead of this, we eventually chose to use deep learning-based object detectors to both detect and classify the honey bees. This option streamlined the workflow and reduced computational burden. As a result, inference time was significantly reduced with optimal performance.

YOLOv10, FCOS, RetinaNet and SSD are one-shot detectors among the models we used in our study while FrCNN is a two-shot detector. FrCNN typically requires longer training and inference time than SSDs, which makes it less suitable for real-time monitoring. While all the models are deep learning-based object detectors, their detection approaches are vastly different and thus well-suited for this study. YOLO predicts bounding boxes and classes directly from the full images. FrCNN proposes regions and then classifies them. RetinaNet uses single pass with focal loss for class imbalance. FCOS directly predicts bounding boxes without a predefined anchor which is present in all other models. SSD discretizes output space into default boxes. These differences make each algorithm better for different purposes. YOLOv10 is mostly used in applications in need of a real-time detection system. FrCNN is used when very high accuracy is needed. RetinaNet is used for applications with large class imbalance and FCOS is used for small object detection applications. These differences in detection methods amplify our study's capacity to detect honey bee activity accurately and efficiently, allowing us to leverage the strengths of each model according to the specific requirement of the tasks. Among the one-shot detectors utilized in this study (YOLOv10, FCOS, RetinaNet, and SSD), performance varied significantly based on the specific architecture. Specifically, FCOS outperformed RetinaNet in the detection task, capturing more honey bees in the frame. Conversely, due to its use of focal loss in training, RetinaNet proved superior at classifying honey bee types.

The high confidence scores observed in FrCNN and YOLOv10, can be attributed to the distinct morphological features of the target classes. For example, the presence of yellow or orange pollen pellets on the hind legs could provide a high-contrast feature that would make the models to differentiate WB Pollen from WB No Pollen class. The larger body size and unique eye structure of drone bees could provide spatial and

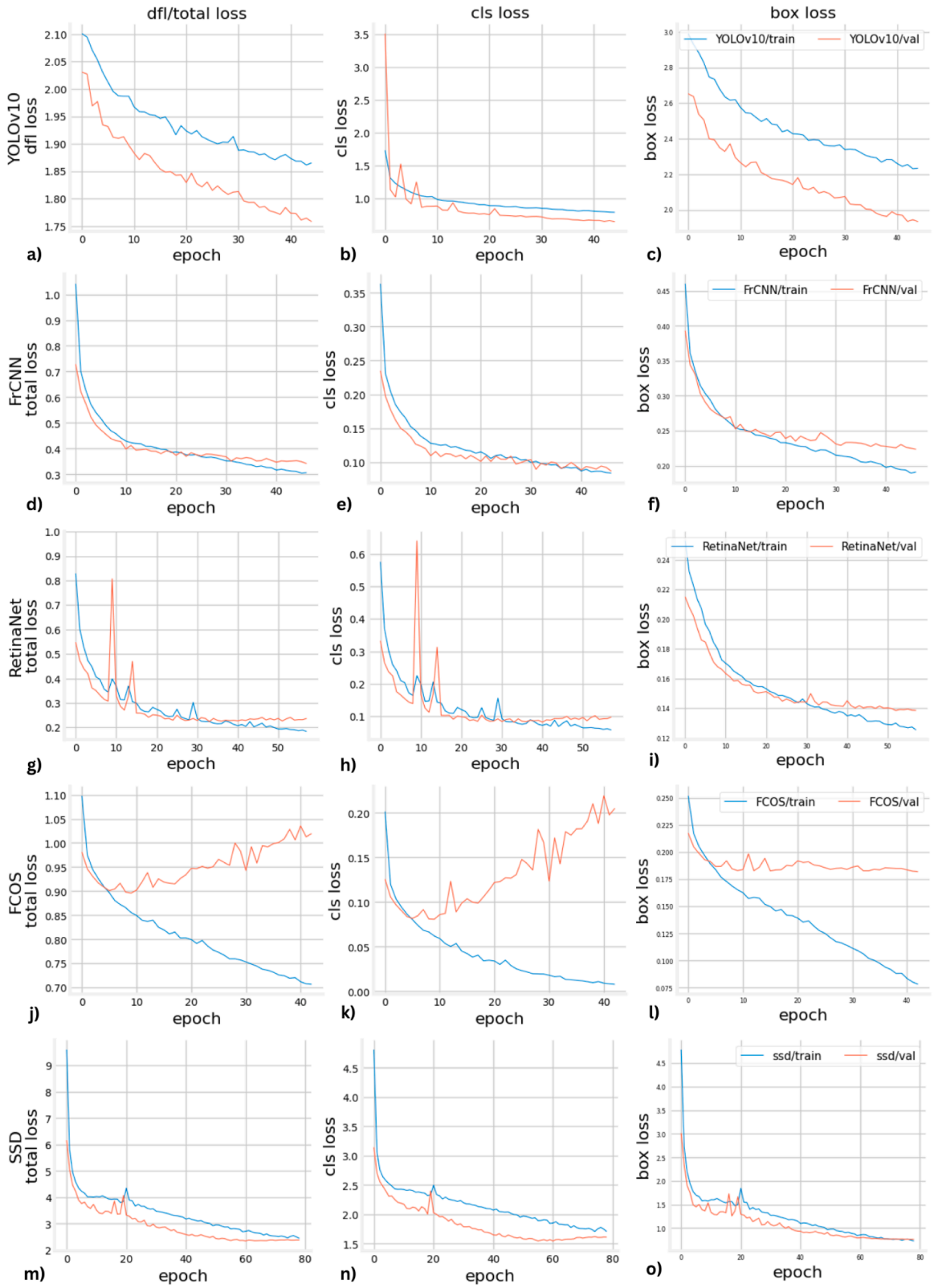


Fig. 3. Training and validation loss for all five trained models: (**a**) dfl loss for YOLOv10, (**b**) cls loss for YOLOv10, (**c**) box loss for YOLOv10, (**d**) total loss for FrCNN, (**e**) cls loss for FrCNN, (**f**) box loss for FrCNN, (**g**) total loss for RetinaNet, (**h**) cls loss for RetinaNet, (**i**) box loss for RetinaNet, (**j**) total loss for FCOS, (**k**) cls loss for FCOS, (**l**) box loss for FCOS, (**m**) total loss for SSD, (**n**) cls loss for SSD and (**o**) box loss for SSD. Box loss, cls loss and dfl loss stand for bounding box loss, classification loss and distributed focal loss, respectively. The x-axis represents the number of epochs, while the y-axis represents the loss value. With ideal training, both training and validation losses should decrease and stabilize over time. If the validation loss diverges from the training loss, it indicates overfitting. Similarly, if both the training and validation loss values are high, it suggests underfitting.

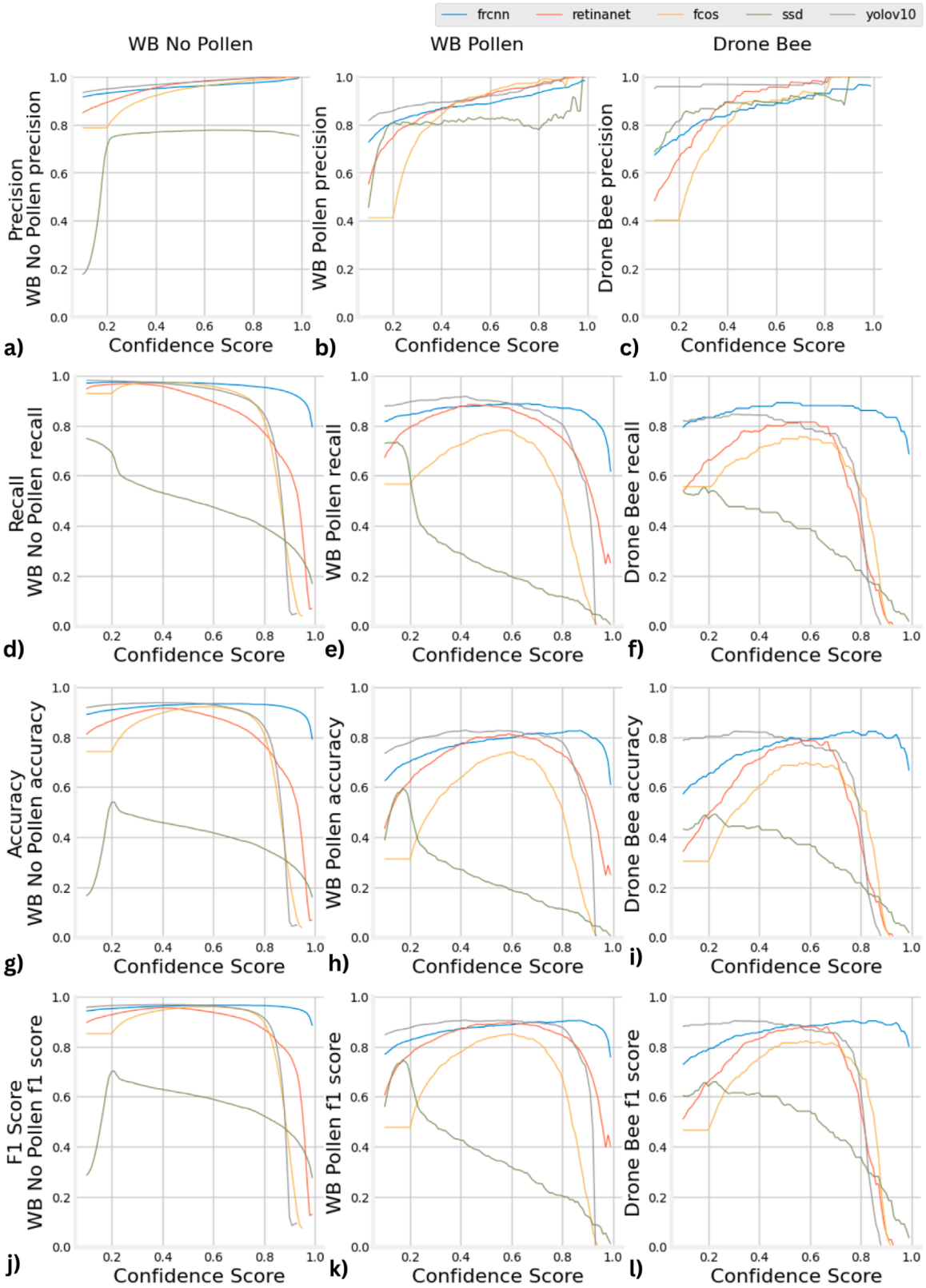


Fig. 4. Class wise (WB no pollen, WB pollen and drone bee) metrics versus confidence score graph of all trained models evaluated on test dataset: (**a**) precision of all models for WB no pollen class, (**b**) precision of all models for WB pollen class, (**c**) precision of all models for drone bee class, (**d**) recall of all models for WB no pollen class, (**e**) recall of all models for WB pollen class, (**f**) recall of all models for drone bee class, (**g**) accuracy of all models for WB no pollen class, (**h**) accuracy of all models for WB pollen class, (**i**) accuracy of all models for drone bee class, (**j**) f1-score of all models for WB no pollen class, (**k**) f1-score of all models for WB pollen class and (**l**) f1-score of all models for drone bee class. WB indicates worker honey bee. The x-axis is confidence score, the threshold where detections made by the model are considered valid. The y-axis represents the evaluation metric. A well-trained model maintains high precision across all confidence level while recall gradually decreases. The F1-scorre is typically a smooth peak where precision and recall balance optimally.

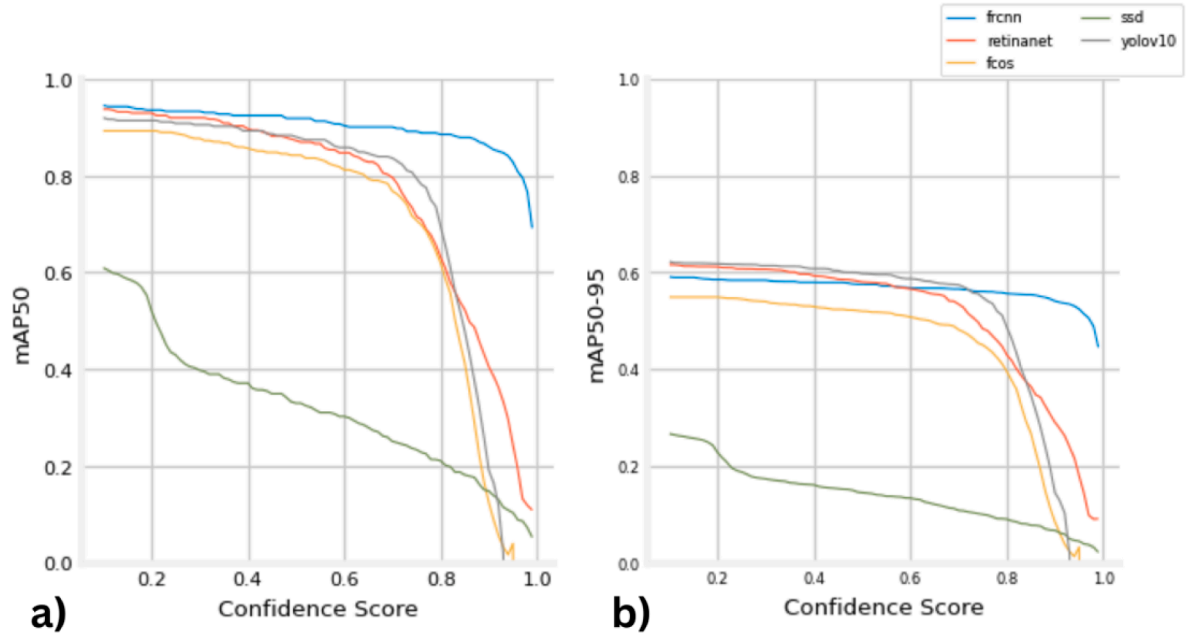


Fig. 5. Mean Average Precision (mAP) score evaluated at different IoU versus confidence score for all models on test dataset: (**a**) mAP@50 score and (**b**) mAP@50-95 score. mAP is a metric used in object detection to evaluate models based on correct labels and precision of object's predicted location and has been described in detail in the method section. The x-axis is confidence score, the threshold where detections made by the model are considered valid. The y-axis represents the mAP50 or mAP50-95 score. A well-trained model maintains high mAP score with smooth decline as confidence score increases.

textural cues that distinguish them from worker bees. These phenotypic differences can reduce class ambiguity during the final Softmax or class-probability calculations, leading to the sustained high-confidence detections observed in our results.

Despite the diversity in our data collection, the dataset created is imbalanced due to the biology of the honey bee colony. To reduce the effects of dataset imbalance, we applied focal loss during training to RetinaNet and FCOS to ensure that less frequent classes contributed more to the learning process. Focal loss assigns more weight to minority classes, reducing bias toward the dominant class. Data augmentation techniques were employed every epoch, such as random crops, mirroring, rotation, and brightness and contrast adjustment, ensuring that the model was exposed to a more diverse dataset. The performance metrics for minority classes are thus high in our results despite the imbalance in data due to the measures taken above, showing its effectiveness.

Our methodology was not comprehensive, and many other object detection models exist. Centernet [60] is a single-stage detector that uses center points of objects and their sizes instead of bounding box proposals. This makes it highly effective when detecting small objects. DETR [61] (DEtection TRansformers) could also be studied for use with honey bees. This model uses transformers [62] for end-to-end detection (unified framework that removes the need for traditional stages of detection like region proposals and subsequent classification). The self-attention mechanism in transformers enables them to focus on certain parts of the input data, which are more important for feature learning. In future, there is also a need to analyze different loss functions for our models, which may further improve performance.

Among the five different models studied, all except the SSD could be deployed in the real-world for bee detection. The class-wise metrics displayed in Fig. 4 highlight that YOLOv10 performed the best overall with FrCNN following closely behind YOLOv10. Notably, FrCNN maintained high recall, accuracy, and F1-score even at high confidence score while the performance of other models started declining at high confidence intervals. It underscores FrCNN's more conservative behavior, relying on high-quality proposals from its trainable region proposal network. FrCNN refined the proposals and examined the regions in detail leading to more confident detections compared to one-shot detectors. In contrast, one-shot detectors produced more false negatives as confidence increased showing the trade-off between speed and detection reliability.

While FrCNN excelled at differentiating between various bee classes, it occasionally struggled to distinguish bees from the background when compared to YOLOv10. However, YOLOv10 struggled with false positives, especially at varying confidence scores. FCOS outperformed RetinaNet in the detection task, capturing more honey bees in the frame. Conversely, due to its use of focal loss in training, RetinaNet proved superior at classifying honey bee types. FrCNN achieves the highest mAP score across all models when calculated at 0.5 IoU (Fig. 5). This indicates its robust performance in detecting and classifying bees with high accuracy. However, when mAP is calculated at a broader range of 0.5 to 0.95 IoU, both YOLOv10 and RetinaNet surpassed FrCNN. This broader range of evaluation highlights the strengths of YOLOv10 and RetinaNet across varying levels of overlap between predicted bounding box and ground truth. A higher mAP score indicates better overall performance, balanced detection across classes and robustness of a model. While mAP scores are widely used in object detection research, mAP@50 provides a good estimate of the model's ability to detect objects with a reasonable overlap threshold and is useful for stakeholders who prioritize high precision and recall. On the other hand, mAP@50-95 gives a more rigorous measure of performance across varying object sizes but is more relevant for model benchmarking than real-world deployment of models. SSD suboptimal performance could possibly be attributed to the scale of the objects. The default anchor boxes in SSD are predefined for a range of aspect ratios and scales. Given that the average size of pollen pellets is 15 by 15 pixels, SSD's default anchor boxes might not be well-suited for such small objects. If these anchors are not well-matched to the size and shape of the honey bees, SSD might not generate accurate proposals for these small objects, especially when the honey bees are densely packed at the honey bee colony entrance.

The primary contributions of this research are threefold (a) the development of a fully non-invasive, robust deep learning system capable of real-time honey bee monitoring without requiring modifications to hive entrances or controlled conditions; (b) the creation and public availability of an annotated dataset of 4,590 frames with 79,212

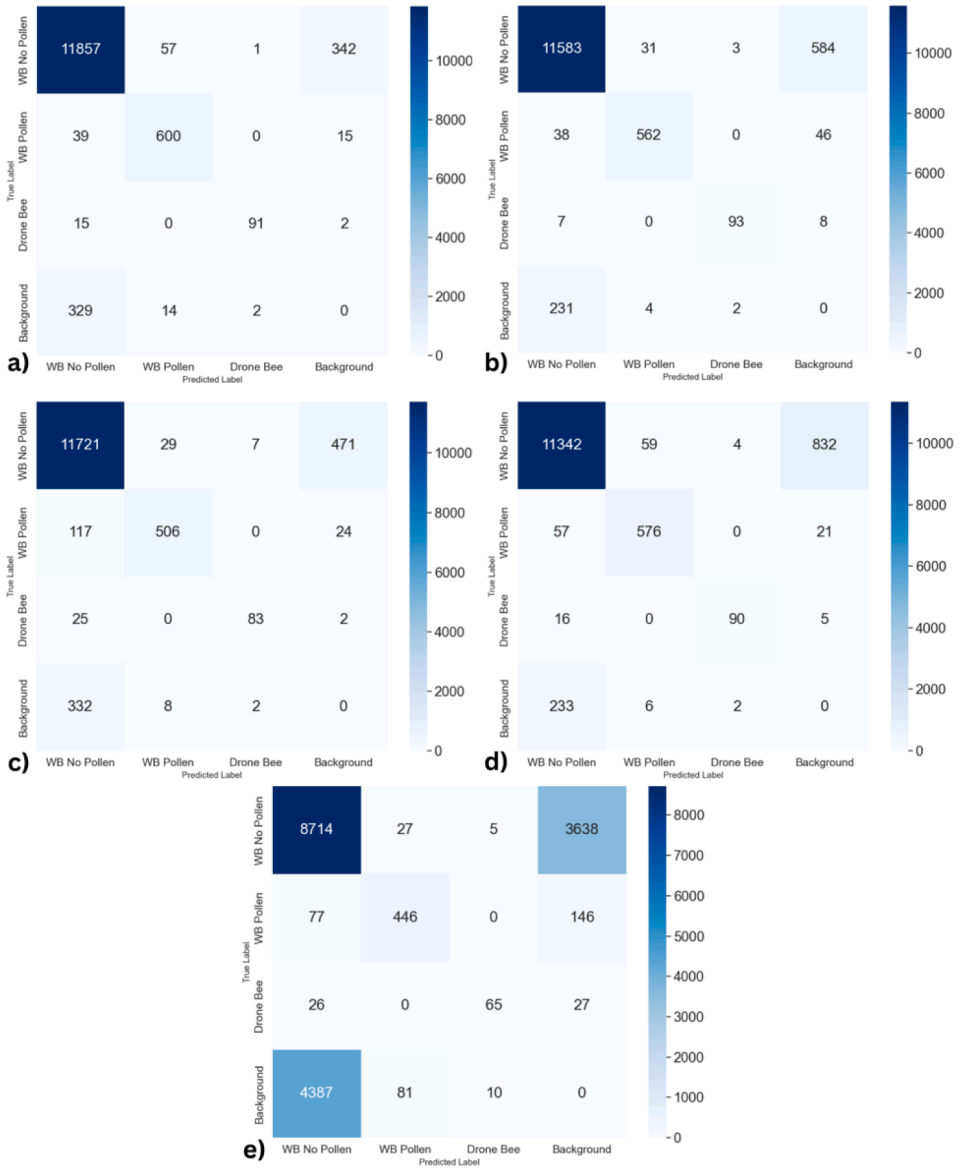


Fig. 6. Confusion matrix of different trained models evaluated on test sets illustrating how often the models misclassified one bee category as another. (**a**) confusion matrix of YOLOv10, (**b**) confusion matrix of FrCNN, (**c**) confusion matrix of FCOS, (**d**) confusion matrix of RetinaNet and (**e**) confusion matrix of SSD. The diagonal values represent correctly classified instances (true positives) for three bee classes. The off-diagonal values represent misclassification. A high count in diagonal cells represents strong classification accuracy. The Background row and column are included to visualize localization errors against the hive entrance environment, where off-diagonal counts indicate false positives and false negatives.

bee annotations, categorized into worker bees with and without pollen, and drone bees; and (c) the study provides a thorough evaluation and comparison of five prominent deep learning object detection models for this specific task, offering insights into their suitability and performance characteristics. This dataset created from this study addresses a gap in existing resources by capturing data under a wide range of natural environmental variations. The statistically significant difference in model performance, confirmed by the Friedman's test, underscores the importance of model selection for this problem.

This research has several theoretical and practical implications. Theoretically, it demonstrates the effectiveness of applying deep learning object detection models to the novel task of automatically identifying and categorizing honey bees at hive entrances in real-world, unmodified conditions. This contributes to the growing body of literature on the use of computer vision and deep learning in entomology and ecological monitoring. The study also highlights the nuances in performance among different model architectures for this specific application, particularly the strong performance of YOLOv10 and FrCNN, and the limitations of SSD for detecting small, densely packed objects like honey bees. Furthermore, the creation and public release of a unique, high-quality dataset comprising two years of video data under diverse environmental conditions provides a valuable resource for the research community, facilitating reproducibility and further advancements in this area.

A key limitation of prior studies was structured setup which disrupted bee behavior and minimized environmental variability. This system offers a non-invasive, cost-effective solution for beekeepers to monitor colony health and assess pollination efficiency. Different state-of-the-art deep learning models and their performances are compared in a real-life scenario to help select what best suits the needs of the beekeepers, researchers and other stakeholders. Our method also incorporates varying environmental conditions as well as light intensity for video recording and this can be easily replicated with minimal time and effort. Traditional methods, such as manual counting or using traps,

can be labor-intensive and may stress the honey bees, potentially impacting their foraging behavior. In contrast, our automated approach minimizes disruption while providing accurate and timely data. From a practical standpoint, the proposed system offers numerous advantages for beekeepers, growers, and researchers. It provides a cost-effective and easy-to-use method for continuously monitoring honey bee activity and potentially assessing colony health and pollination efficiency without disturbing the hives. The use of readily available GoPro cameras and the potential for deployment with GPU-embedded hardware like the NVIDIA Jetson enable in-situ processing and real-time insights into colony foraging dynamics.

Many beekeepers may not have the computational resources or technical expertise to run deep learning models locally. A website or application could be created for users to upload their recorded videos and inference results can be returned to them via the same platform. These setups will allow any user (stakeholders such as beekeepers and growers) to receive immediate insight into colony foraging activity and health, facilitating informed decision-making while ensuring minimal interference with the honey bees' natural behavior. A website or application-based system eliminates the need for high-end hardware or software installations. They only need to record the beehive videos and upload it to the website to get inference results. For practical deployment, not just a web-based application, hardware costs must also be considered. A GoPro camera costs $250-$350, which can be substituted with a cheaper camera. An NVIDIA Jetson Orin nano for real-time inference costs $250-$300 which can be upgraded for improved performance. Maintenance primarily involves periodic lens cleaning, camera positioning adjustments, and power supply management. The system used in our study is designed for minimal technical expertise. The user can set the camera and microprocessor on beehive entrances with ease, connect the camera with the microcomputer and run the script to start inferring results, making it accessible for beekeepers and researchers.

Despite the strengths of this research, there are several limitations to acknowledge. The dataset, while extensive, exhibits an imbalance in the representation of different bee classes, particularly worker bees with pollen and drone bees, which could potentially affect the training and performance evaluation of the models on these less frequent categories. Furthermore, the annotation process, while guided by expert knowledge and subject to multiple reviews, did not involve formal inter-annotator agreement metrics due to the scale of the dataset and limited resources. While the study considered varying lighting and weather conditions, the specific range and intensity of these conditions were limited to the data collected in a particular geographical location over a two-year period. The dataset was generated exclusively from the research apiary at Mississippi State University. Consequently, the models were trained on the specific phenotypic characteristics of the local *Apis mellifera* sub-species and within the constraints of standard honey bee colony boxes. Variations in bee morphology (differences in coloration and size, across various sub-species) could impact the accuracy of caste classification and pollen detection. The model's generalization may be influenced by hive architecture, different entrance configurations, landing board materials, or background textures could introduce novel visual noise not represented in our current dataset. While the system demonstrates high reliability in the captured environment, further validation across diverse apiary sites, hive designs, and honey bee lineages is required to ensure broad ecological and commercial applicability. A detailed computational complexity analysis of the models for real-time deployment across different hardware platforms was also beyond the scope of this initial study.

Future research could build upon this work in several promising directions. Investigating hybrid deep learning models, such as transformer-based detection architectures, could potentially lead to improved feature learning and enhanced performance, particularly in handling occluded or rapidly moving bees. Exploring the application of data augmentation techniques that specifically simulate more challenging environmental conditions, such as heavy rain, fog, or extreme temperatures, could further enhance the robustness and real-world applicability of the system. Using models which reduce or avoid high false positive rates will also ensure more accurate estimates of pollen foragers and colony conditions for practical applications in beekeeping and pollination experiments. Future studies could focus on developing a user-friendly software platform or mobile application that integrates this deep learning system, allowing beekeepers and researchers to easily collect, upload, and analyze video data from their beehives, thereby facilitating wider adoption and impact of this technology. Recently there has been an increasingly unprecedented honey bee colony losses in the United States [63]. It is of utmost importance for beekeepers to be able to continuously monitor colonies as needed. Our study is thus unique and provides the first step towards building a non-invasive, cost effective and easy platform to help stakeholders and researchers monitor honey bee colonies.

5. Conclusion

This study demonstrates that deep learning-based computer vision models can accurately and non-invasively monitor honey bee colonies in real-world conditions. By developing a robust dataset of 4,590 frames encompassing diverse weather and lighting environments, we evaluated five state-of-the-art object detection architectures (YOLOv10, Faster R-CNN, RetinaNet, FCOS, and SSD) across multiple performance metrics. YOLOv10 and Faster R-CNN consistently achieved the highest accuracy and F1 scores, indicating their suitability for real-time forager detection and classification. The statistically significant differences identified through Friedman's test confirm the importance of model choice in apicultural monitoring applications.

The proposed system offers a scalable, low-cost, and non-intrusive alternative to traditional manual or sensor-based colony strength assessments. The freely available annotated dataset and open methodology facilitate reproducibility and enable further research on automated pollination monitoring. While the dataset is inherently imbalanced due to natural caste distributions, and annotations were limited by expert capacity, the models nonetheless performed robustly across environmental conditions. Future work should focus on expanding the dataset across additional geographic regions, integrating transformer-based architectures for improved small-object detection, and developing user-friendly mobile or web-based inference interfaces to enable widespread adoption by beekeepers and researchers.

Data availability statement

Raw data was generated at the research apiary on the campus of Mississippi State University. The labeled dataset along with other files that support the findings of this study are openly available at Scholars Junction at https://doi.org/10.54718/JCHW9401. The data can be accessed for reproduction and further exploration of beehive monitoring. Raw video recordings of bee hives can be made available upon reasonable request to the corresponding authors (Sathish Samiappan and Priyadarshini Chakrabarti).

Ethics in publishing statement

I testify on behalf of all co-authors that our article submitted followed ethical principles in publishing.

All authors agree that:

This research presents an accurate account of the work performed, all data presented are accurate and methodologies detailed enough to permit others to replicate the work.

This manuscript represents entirely original works and or if work and/or words of others have been used, that this has been appropriately cited or quoted and permission has been obtained where necessary.

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All authors have been personally and actively involved in substantive work leading to the manuscript and will hold themselves jointly and individually responsible for its content.

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CRediT authorship contribution statement

Piyush Chaudhary: Writing – original draft, Validation, Software, Formal analysis. **Connor Foley:** Writing – original draft, Formal analysis, Data curation. **Sathishkumar Samiappan:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization. **Leon Kohler:** Data curation. **Volkan Senyurek:** Writing – review & editing, Investigation, Funding acquisition. **Dawson Boes:** Validation, Data curation. **Priyadarshini Chakrabarti:** Writing – review & editing, Supervision, Resources, Project administration, Investigation, Funding acquisition, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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